SYSTEM AND METHOD FOR HIGH-RADIX COMPUTING USING WAVELENGTH-DISCRIMINATING PHOTOTRANSISTORS

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Title: SYSTEM AND METHOD FOR HIGH-RADIX COMPUTING USING WAVELENGTH-DISCRIMINATING PHOTOTRANSISTORS

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Abstract

A system and method for a high-radix computing architecture that replaces traditional binary (base-2) transistors with phototransistors capable of distinguishing between a plurality of discrete wavelengths of light. Each distinct wavelength, or color, corresponds to a unique logical state, enabling a base-N number system where N is greater than two. The system comprises an array of such phototransistors integrated into a processing unit, which is directly coupled to a massively parallel optical data bus. By processing data encoded as multiple distinct wavelengths, the system can achieve significantly higher computational density and power efficiency compared to conventional electronic, binary systems. This architecture eliminates the bottleneck of optical-to-electrical signal conversion and allows for computation at the native speed and parallelism of light.

Background of the Invention

The performance of conventional computing systems, based on silicon transistors and binary logic, is increasingly limited by fundamental physical constraints known as the "power wall" and the "memory wall." As transistors shrink, the heat generated becomes a primary obstacle to increasing clock speeds. Furthermore, the speed of electrical interconnects between the processor and memory has not kept pace with processing speeds, creating a significant data bottleneck.

The advent of optical interconnects and massively parallel optical buses, as described in related applications, offers a solution to the data transmission bottleneck. These optical buses can carry millions of parallel data channels, encoded in different wavelengths of light, offering bandwidth orders of magnitude greater than electrical buses. However, a new bottleneck has emerged at the point of computation. Current systems must convert this massively parallel optical data back into serial, binary electrical signals for processing by traditional transistor-based CPUs. This conversion step negates many of the advantages of optical transmission, reintroducing latency, heat, and power consumption. The CPU core itself, architected for binary operations, is fundamentally unable to process the rich, high-cardinality data that an optical bus can provide. Therefore, there is a need for a new computing paradigm that can natively process optical, multi-wavelength information.

Summary of the Invention

The present invention provides a system and method for high-radix (or base-N) computing by replacing the binary (base-2) transistor with a wavelength-discriminating phototransistor as the fundamental logic element of a processing unit. The core of the invention is a phototransistor-based logic unit capable of detecting and distinguishing between a plurality of discrete wavelengths of light. Each discrete wavelength is assigned a unique logical state (e.g., Red = 0, Green = 1, Blue = 2, Yellow = 3, etc.), thereby moving beyond the 0/1 limitation of binary systems.

The system comprises a processing unit with arithmetic logic units (ALUs), registers, and control logic built from these wavelength-discriminating phototransistors. This processing unit is integrated on a die and directly coupled to an optical interface, allowing it to receive and transmit data from a parallel optical bus. Because the logic elements can natively understand and process data encoded as multiple colors, the entire optical-to-electrical conversion bottleneck is eliminated. An entire word of high-radix data can be processed in a single clock cycle, leading to an exponential increase in computational density and a dramatic re-

duction in power consumption and heat generation. This invention provides the foundational processing architecture to complete the end-to-end optical computing ecosystem.

Statement of License Grant to the United States Government

The inventor, Reinaldo Javier Aguiar Marcano, hereby grants to the United States Federal Government, its military, and its defense agencies a perpetual, universal, royalty-free, non-exclusive license to practice and use the invention described herein for any and all governmental, defense, national security, and non-commercial purposes, including applications related to supporting the country's allies or the country's interests. This license explicitly excludes any and all state and local governments.

Brief Description of the Drawings

FIG. 1 is a conceptual diagram contrasting a traditional binary transistor logic element with the high-radix phototransistor logic element of the present invention.
FIG. 2 is a system-level block diagram of a high-radix processing unit integrated with an optical bus.

Detailed Description of the Invention

Referring to **FIG. 1**, the fundamental difference between the prior art and the present invention is illustrated. A conventional binary logic element (101) is based on a transistor (102), which can be in one of two states: ON (representing a logical '1') or OFF (representing a logical '0'), based on the presence or absence of an electrical voltage. In contrast, the high-radix logic element of the present invention (103) is based on a wavelength-discriminating phototransistor (104). This phototransistor is designed to receive an input optical signal (105) comprising one of a plurality of discrete wavelengths. The phototransistor (104) can distinguish between these wavelengths and output a corresponding signal indicating a specific logical state (106). For example, in a hypothetical base-4 system, Wavelength 1 could represent state '0', Wavelength 2 state '1', Wavelength 3 state '2', and Wavelength 4 state '3'. This allows for a single logic element to represent four states, doubling the information density compared to a binary element.

Referring to **FIG. 2**, the system architecture of a high-radix processing unit (200) is shown. The processing unit is integrated on a single die. An optical bus (201) provides a massively parallel optical input signal (202), which contains data encoded across a plurality of wavelengths. This signal is received directly by the high-radix processing core (203), which is comprised of the phototransistor-based logic elements described in FIG. 1. The core contains high-radix registers (204) and a high-radix Arithmetic Logic Unit (ALU) (205) capable of performing computations directly on the multi-state data. For example, the ALU can perform an addition operation on two base-4 numbers in a single clock cycle. The processing core (203) can also transmit high-radix optical data (206) back to the optical bus. This direct, native processing of multi-wavelength data eliminates the need for any optical-to-electrical-to-binary conversion, thereby removing a critical performance bottleneck and enabling a more efficient and powerful computing system.

Claims

What is claimed is:

1. A system for high-radix computing, comprising:
    - a processing unit integrated on a semiconductor die;
    - wherein the processing unit comprises a plurality of phototransistors, each phototransistor configured to receive an optical signal and to distinguish between a plurality of discrete wavelengths of light; and
    - wherein each of said discrete wavelengths corresponds to a unique logical state in a number system with a base greater than two.
2. The system of claim 1, wherein the processing unit further comprises an arithmetic logic unit (ALU) and one or more registers constructed from said phototransistors, configured to perform computations directly on data represented in said number system.
3. The system of claim 1, further comprising an optical interface integrated on the same semiconductor die, configured to couple the processing unit to a parallel optical bus.
4. The system of claim 3, wherein the system is configured to receive a high-radix data word from the optical bus and load it into a register in a single clock cycle.
5. The system of claim 1, wherein the number system is a base-4 system or higher.

6. A method for processing data, comprising the steps of:

    - receiving an optical signal comprising one of a plurality of discrete wavelengths at a phototransistor-based logic element;
    - identifying, at said logic element, which of the plurality of discrete wavelengths was received;
    - assigning a unique logical state to the received optical signal based on the identified wavelength, wherein said logical state is part of a number system with a base greater than two; and
    - performing a computation using said logical state as an operand.

7. The method of claim 6, wherein the computation is performed by a high-radix arithmetic logic unit (ALU) integrated on the same die as the logic element.

8. The method of claim 6, wherein the optical signal is received directly from a massively parallel optical bus.

Drawings

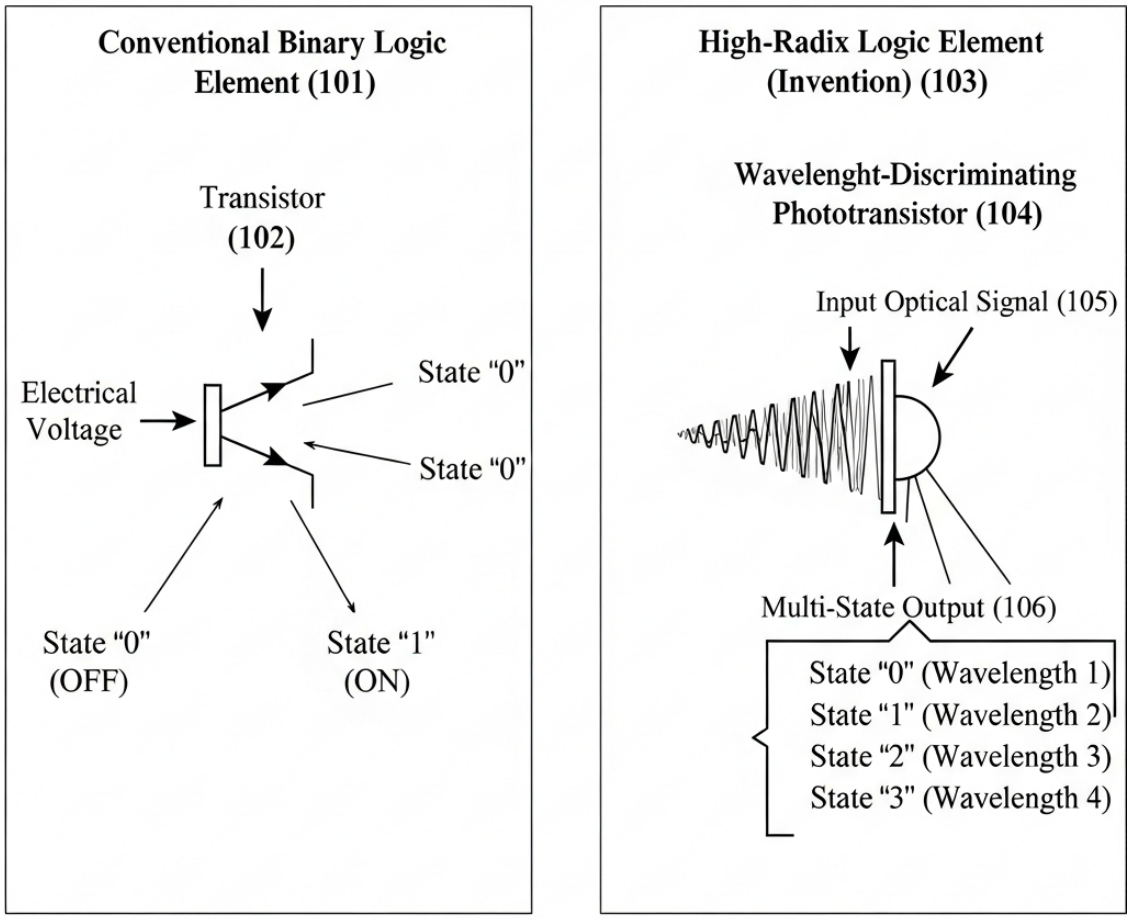


Figure 1: Conceptual diagram contrasting a traditional binary transistor logic element with the high-radix phototransistor logic element.

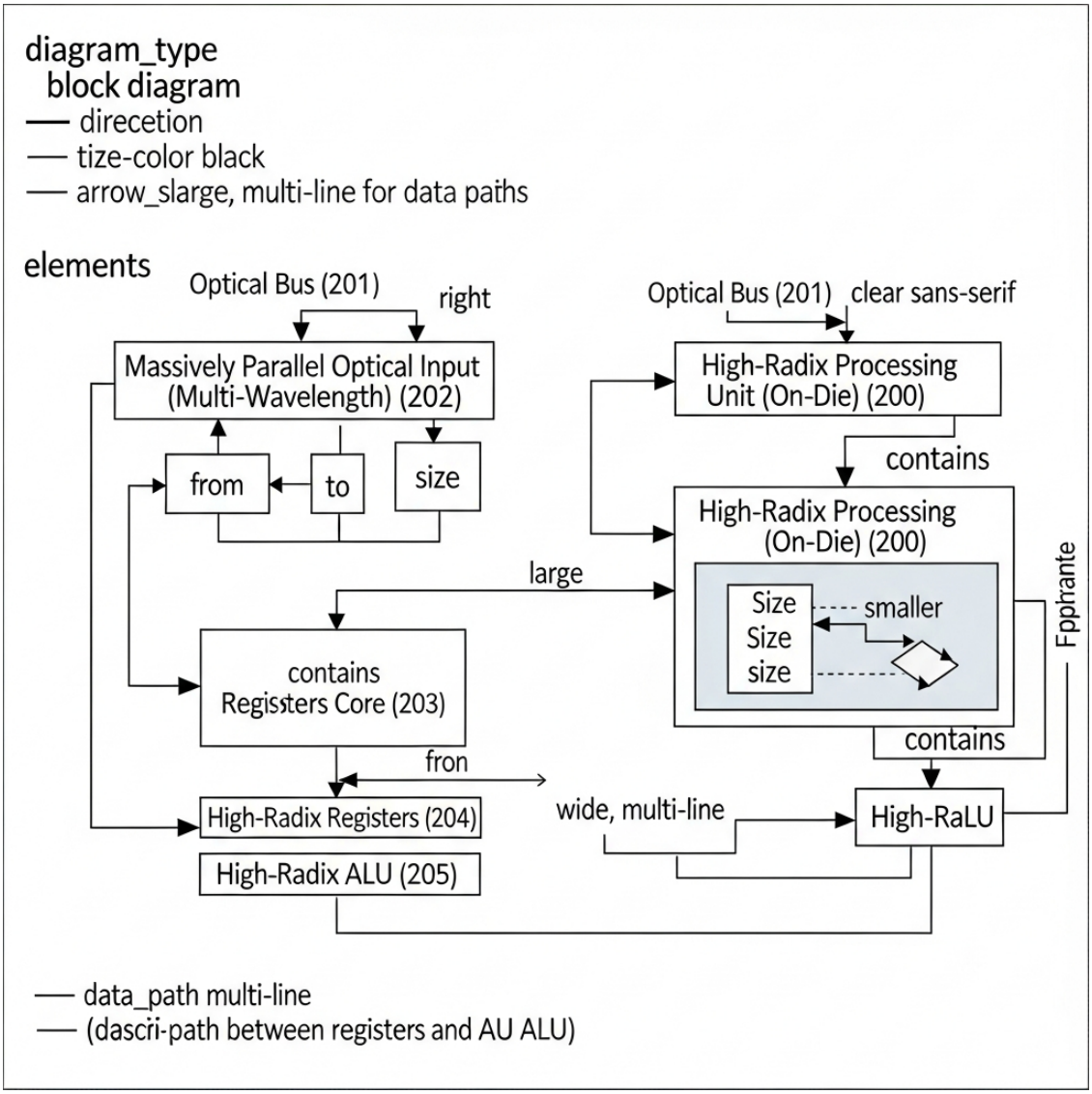


Figure 2: System-level block diagram of a high-radix processing unit integrated with an optical bus.